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Mohamed Hamed

AI Full-Stack Engineer · Agentic Systems, RAG & MCP · Product Owner (PSPO I)

SUMMARY

AI full-stack engineer who also owns the product. I build multi-agent systems, MCP servers, RAG pipelines with embeddings and knowledge graphs, and local, sovereign AI that keeps data on-device, then ship them inside production web and mobile apps. Ten public repositories show the work end to end: a 7-agent team that turns Arabic or English Telegram requests into GitLab merge requests, a LangGraph agent office, a local RAG workspace with hybrid retrieval and cited answers, and the evaluation corpus used to test it. As a PSPO I certified Product Owner at Bonyad I run four products (combined budgets above 1,000,000 SAR) with teams in Egypt, the UAE and Saudi Arabia, so I start from the metric, scope the smallest useful slice and measure it after release. Native Arabic speaker, RTL-first.

TECHNICAL SKILLS

Agents	Multi-agent orchestration (orchestrator / planner / workers / reviewer), LangGraph StateGraph, LangChain ReAct tool calling, handoffs, human-in-the-loop approvals, sandboxed computer-use and GitHub/GitLab tools, structured outputs with validated action plans
RAG & retrieval	Embeddings (Ollama qwen3-embedding), chunking with stable statement anchors, hybrid search (SQLite FTS5 + vector + graph, reciprocal rank fusion), citations and honest refusal, temporal "as-of" queries, LLM entity/relation extraction, knowledge graphs, OCR ingestion (PDF, DOCX, images)
MCP, harnesses, skills	Built 3 MCP servers in TypeScript (MCP SDK); wired 35+ standard MCP servers per agent; Claude Code (skills, subagents, hooks, MCP); DeepSeek Harness plugin authoring (agent tools + UI panels); prompt design, context engineering
Models & LLMOps	Anthropic Claude, OpenAI GPT-4o / Whisper, Zhipu GLM, Moonshot Kimi, DeepSeek, Ollama (local), whisper.cpp; per-agent provider routing, cost tracking, eval sets with held-out gold answers, deterministic dataset generators and validators
Full-stack	TypeScript, JavaScript, Python, Swift, Kotlin · Next.js (App Router, server actions), React, Tailwind, Node.js/Express, FastAPI, Socket.io, WebSockets, Server-Sent Events streaming, REST · Three.js
Data & cloud	PostgreSQL (Neon, PL/pgSQL), MySQL, SQLite/FTS5 · Docker, AWS EC2, PM2, Vercel (Blob, Firewall), GitHub Actions, GitLab CI, fastlane/TestFlight · rate limiting, JWT refresh flows, Keychain
Mobile	SwiftUI (MVVM, async/await, actors), watchOS + HealthKit, AVKit/HLS, Kotlin + Jetpack Compose, Firebase, RTL and i18n (Arabic, English, French, Spanish)
Product	Roadmaps, backlog ordering by value, Evidence-Based Management, KPIs (activation, retention, conversion), PRDs, user stories, UAT, Scrum and Kanban, stakeholder management

AI ENGINEERING PROJECTS — github.com/MohamedHamed163

Nigents — self-hosted multi-agent dev team for GitLab FLAGSHIP 2026

Stack: Node.js, Express, Socket.io, MySQL, Anthropic/OpenAI SDKs, GLM, Kimi, DeepSeek, MCP SDK, Whisper, GitLab API, Telegram, Docker, PM2, AWS EC2

- Built a 7-agent team (Orchestrator, Planner, Backend, Frontend, QA, Code Reviewer, Reporter) that turns a Telegram text or Arabic/English voice note into a plan, a human-approved implementation on a task branch and a GitLab merge request.
- Wrote 3 custom MCP servers (Arabic RTL auditor, task splitter, semantic code search) and connected 35+ standard servers, for 38 tools; agents may only edit files listed in the approved plan.
- Live dashboard over Socket.io with a drag-and-drop Workflow Studio, model stats, per-agent provider choice across 5 LLM vendors and per-task cost tracking; multi-user with admin panel; runs 24/7 on AWS EC2.

NotchWiki — local RAG and knowledge-graph workspace (macOS) 2026

Stack: Swift, SwiftUI, SQLite FTS5, Ollama (qwen3.5, qwen3-embedding), Vision OCR, vDSP, Python sidecar

- Hybrid retrieval merging FTS5 keyword, vector and graph-neighbourhood results with reciprocal rank fusion; every answer cites the exact source statement or refuses when there is no evidence.
- Ingestion for Markdown, PDF (incl. scanned via on-device OCR), DOCX and images with stable statement anchors; embedding-model versioning that marks vectors stale instead of mixing models; "as-of-date" answers that report superseded facts.
- LLM entity/relation extraction with review states, generated wiki pages with staleness tracking, and a local Zep-compatible memory service for agent simulations. 100% local inference; 25 unit tests.

Aster Vale — RAG evaluation corpus 2026

- Deterministic generator and validator for 2,000 documents across 100 projects with a ground-truth graph (every fact quoted to source), 200 held-out questions with gold answers, 240 personas and 4 counterfactual scenarios; used to test NotchWiki at 20 / 200 / 2,000-document scales.

Agent Room — LangGraph agent team in a live 3D office 2026

Stack: Python, FastAPI, WebSockets, LangGraph, LangChain, GPT-4o, GLM-4.6, Three.js

- Supervisor StateGraph with 5 ReAct tool-calling agents (research, coding, GitHub PRs, computer use); callback handlers stream thinking/tool events that drive the 3D scene; approve/deny gates for every risky action and a PR-only GitHub policy.

Wake Mac — on-device voice agent · DeepSeek Harness plugins — agent tools 2026

- Wake Mac: wake phrase → whisper.cpp → local Ollama planner → *PlanValidator* that checks each step against a closed action schema before execution; nothing leaves the Mac; 21 tests.
- Harness plugins: a project tracker whose UI and agent tools (*issue_board, issue_write, project_write*) share one store, plus a theme and tab bar; 8 locales, test runners per plugin.

EXPERIENCE

Product Owner & AI Full-Stack Builder — Bonyad 2026 – Present · Remote (Egypt, UAE, KSA)

Four web and mobile products · combined budgets above 1,000,000 SAR

- **Droob** (AI learning platform, live on iOS, Saudi Arabia): owned the product from idea to launch and built the native SwiftUI student app, including an AI tutor that streams answers with sources over Server-Sent Events, HLS lessons, and full Arabic RTL.
- **Mafeshsho8l** (AI job-search, live on iOS and Android): took it from an empty backlog to both stores; defined the AI CV builder and AI skills-gap roadmap from job-seeker research.
- **QuickIn** (vacation rentals, Egypt): turned a vague "booking website" request into a backlog, then built the SwiftUI and Kotlin/Compose apps on a shared Next.js API and Neon Postgres, with CI to TestFlight and APK. **Tawazon**: iPhone + Apple Watch coaching app (HealthKit, STOMP chat, Stripe).
- Set activation, conversion and retention goals per product, reviewed them after every release and re-ordered the backlog; prototyped flows with Claude Code before engineering, cutting hand-off ambiguity.

Full-Stack Engineer (freelance) — Skyline real estate, Egypt 2026

Stack: Next.js App Router, TypeScript, Tailwind, Neon Postgres, Vercel Blob, Vitest

- Built the listing platform and Arabic/English admin: broker accounts, approval workflow with held edits and diffs, leads, rental calendar with weekend and seasonal pricing, audited admin impersonation, per-IP rate limiting and cache invalidation; one deployment served on two domains.

Product Owner — Juleative 2025 – 2026 · Remote (UAE, Egypt)

- Led the rebuild of Hub Drive, a legacy car-rental platform, into a marketplace (ASP.NET Core, React, TypeScript, admin panel, driver app) launched on the App Store, using a three-phase release plan presented to executives.
- Ran client discovery and turned needs into a backlog the team started within a week; wrote proposals, SOWs and PRDs for several client products in parallel.

CERTIFICATIONS

Professional Scrum Product Owner I (PSPO I) — Scrum.org, June 2026 (cert. 1331062)

PSPO II training (Advanced Product Owner) — Scrum.org: Evidence-Based Management, product strategy

Foundations of Project Management — Coursera, Dec 2025

Fundamentals of UI/UX Design — Coursera, May 2024

EDUCATION & LANGUAGES

B.Sc. Business Informatics — German University in Cairo (GUC), Oct 2024; semester abroad at GIU Berlin

Arabic — native · **English** — professional working proficiency

Built Arabic-first AI: Arabic voice commands, RTL auditing MCP server, bilingual apps across KSA, UAE and Egypt.